ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Section A: objective questions (30) · Section B: pneumatics and electro-pneumatics, hydraulics, electric motors and drives, sensors and transducers, mechanical drives and maintenance (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Pascal's law states that pressure applied to a confined fluid …

A decreases with distance. B acts only downwards.
C turns into heat. D is transmitted equally in all directions.

1.1.2 A double-acting cylinder …

A uses a spring to retract. B uses air to extend and to retract.
C cannot be reversed. D works only with oil.

1.1.3 A 5/2 valve has …

A two ports and five positions. B five positions and five ports.
C two ports and two positions. D five ports and two positions.

1.1.4 The component that removes water and dirt from compressed air is the …

A silencer. B filter (service unit).
C accumulator. D flow control valve.

1.1.5 A solenoid valve is operated by …

A a foot pedal only. B water pressure.
C an electromagnet. D a spring only.

1.1.6 The speed of an AC induction motor depends mainly on the …

A supply frequency and number of poles. B size of the shaft.
C colour of the wires. D length of the cable.

1.1.7 A device that changes the speed of an AC motor by changing the frequency is a …

A transformer. B variable speed drive (VSD).
C rectifier only. D relay.

1.1.8 A sensor that detects metal objects without contact is a/an …

A capacitive humidity sensor. B LDR.
C microphone. D inductive proximity sensor.

1.1.9 The hydraulic component that stores fluid under pressure is the …

A accumulator. B reservoir only. C filter. D cylinder.

1.1.10 A flow control valve on a cylinder is used to control its …

A direction. B pressure only. C speed. D temperature.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Hydraulic systems can produce much larger forces than pneumatic systems of the same size.

1.2.2 Air is incompressible.

1.2.3 A pressure of 1 bar is equal to 100 kPa.

1.2.4 A reed switch on a cylinder is operated by a magnet in the piston.

1.2.5 Reversing two of the three supply phases reverses a three-phase motor.

1.2.6 A capacitive sensor can detect non-metal objects such as plastic or liquids.

1.2.7 Hydraulic oil leaks are harmless to the environment.

1.2.8 A thermocouple measures temperature.

1.2.9 A timing belt has teeth so that it does not slip.

1.2.10 Lock-out procedures are not needed when working on pneumatic systems.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Force per unit area	A	load cell
1.3.2	Reduces the noise of exhaust air	B	pressure
1.3.3	Allows flow in one direction only	C	silencer
1.3.4	Gives an output only when both inputs have air (pneumatic AND)	D	shuttle valve
1.3.5	Gives an output when either input has air (pneumatic OR)	E	transducer
1.3.6	Limits the maximum pressure in a hydraulic system	F	pressure relief valve
1.3.7	Measures force or weight using strain gauges	G	contactor with overload relay
1.3.8	Motor that runs on DC and uses brushes and a commutator	H	non-return (check) valve
1.3.9	Starts and stops a motor and protects it from overload	I	brushed DC motor

1.3.10	Converts a physical quantity into an electrical signal	J	two-pressure valve

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: PNEUMATICS AND ELECTRO-PNEUMATICS

2.1 Name FIVE components of a compressed-air supply system, from the compressor to the cylinder, and give the function of each. (10)

2.2 A double-acting cylinder has a bore of 50 mm and works at 600 kPa. Calculate the theoretical extending force. (4)

2.3 A cylinder must extend only when an operator presses two buttons at the same time (two-hand safety control). Name the pneumatic valve used and explain why this is safer. (3)

2.4 Explain how a 5/2 solenoid valve and two reed switches can make a double-acting cylinder extend and retract automatically. (5)

[22]

QUESTION 3: HYDRAULICS

3.1 In a hydraulic press, the small piston has a diameter of 20 mm and the large piston a diameter of 100 mm. A force of 200 N is applied to the small piston.

3.1.1 Calculate the pressure in the system. (4)

3.1.2 Calculate the force produced by the large piston. (3)

3.1.3 Explain why the large piston moves a shorter distance than the small piston. (2)

3.2 Give THREE advantages and TWO disadvantages of hydraulic systems. (5)

[14]

QUESTION 4: MOTORS AND DRIVES

4.1 Calculate the synchronous speed of a 4-pole AC induction motor on a 50 Hz supply. Use $n = 120f \div p$. (3)

4.2 The motor in QUESTION 4.1 runs at 1 440 r/min at full load. Calculate the slip as a percentage. (3)

4.3 Explain the purpose of the start, stop and overload (thermal) components in a direct-on-line (DOL) motor starter. (4)

4.4 Give TWO advantages of using a variable speed drive (VSD) on a conveyor motor. (2)

4.5 A motor delivers a torque of 15 N·m at 1 440 r/min. Calculate the output power. Use $P = 2\pi nT \div 60$. (3)

[15]

QUESTION 5: SENSORS, MECHANICAL DRIVES AND MAINTENANCE

5.1 Recommend a suitable sensor for EACH and explain why: (a) detecting cardboard boxes on a conveyor; (b) measuring the level of water in a tank without contact; (c) checking the mass of bags of maize meal. (6)

5.2 Compare a belt drive and a chain drive under the headings slip, noise and maintenance. (6)

5.3 Name SEVEN routine maintenance checks for an electro-pneumatic machine. (7)

[19]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER